The Reductionist Betrayal: A Systems Biology Audit of the Pharmaceutical-Industrial Complex and the Suppression of Botanical Synergy

The structural architecture of modern pharmacology is predicated on a reductionist paradigm that prioritizes the isolation, synthesis, and patenting of single bioactive molecules. This approach, while facilitating the standardization of chemical interventions, often disregards the profound biological wisdom inherent in complex plant matrices. For those who view this shift as a betrayal of the co-evolved biological contract between humans and the natural world, the stripping of a plant of its synergistic compounds represents more than just a scientific preference; it is a systematic privatization of nature for profit.[1] This report examines the divergence between the current pharmaceutical "death machine"—characterized by concentrated, often toxic isolates—and the "life strategies" offered by holistic plant complexes. By auditing the clinical omission of non-patentable synergies and the endogenous architecture of human homeostasis, this analysis exposes the "systemic violence" of a medical model that monetizes suffocation and enforces biological fragility.[3]

The Philosophy of Isolation: Reductionism as a Financial Imperative

The pharmaceutical industry's reliance on reductionism is not merely a consequence of scientific progress but a byproduct of a regulatory and legal framework that necessitates exclusivity for profitability. In the prevailing model, the Food and Drug Administration (FDA) clinical trial structure requires a single, stable molecule to prove "efficacy" against a specific biological target.[5] This one-to-one mapping between molecule and receptor is designed to facilitate the creation of Intellectual Property (IP), allowing corporations to claim exclusive ownership over chemical structures for extended periods.[7]

The Intellectual Property Barrier to Synergy

A whole plant cannot be patented. Consequently, the industry has no financial incentive to study the complex herbal groupings that have been shared among organisms for millennia. The resulting "death machine" cycle begins by "mining" a plant for its most potent component, followed by its synthesis and concentration to levels that often exceed natural safety thresholds.[1] When these high-dose isolates produce adverse effects—such as the adrenal suppression caused by synthetic corticosteroids—the system responds by prescribing a second synthetic drug to treat the symptoms of the first.[3] This creates a state of "enforced

fragility," where the organism becomes biologically dependent on a series of synthetic "plug-ins" rather than a self-correcting system designed for homeostasis.[11]

Patent Classification	Focus of Protection	Strategic Implication
Primary Patent	Novel Active Pharmaceutical Ingredient (API)	Establishes 20-year market monopoly [7]
Secondary Patent	Novel formulations, ratios, or dosages	Extends exclusivity through "lifecycle management" [7]
Tertiary Patent	Delivery devices (e.g., inhaler mechanisms)	Blocks generic entry through device-only claims [4]

The prevalence of tertiary patents is particularly notable in respiratory medicine. A cohort study of 331 drug-device combinations approved between 1986 and 2023 revealed that 54% of listed patents were on the delivery devices rather than the active ingredients.[4] This strategy effectively "gatekeeps the breath" of the population, maintaining high markups on essential medications by exploiting device monopolies long after the original drug patents have expired.[4]

The Power of "Entourage Effects": Systems Biology vs. Chemical Hijack

In a holistic plant strategy, the various constituents work together in a way an isolated synthetic drug cannot. This phenomenon, known as the "entourage effect," suggests that the whole plant matters more than the sum of its parts.[13] When cannabis is experienced as a whole plant rather than isolated fragments, the relationships between cannabinoids, terpenes, and flavonoids remain intact, shaping the overall effect in ways that isolated THC or CBD cannot achieve.[14]

Buffer Compounds: The Evolutionary Safety Net

Natural sources often contain secondary chemicals that mitigate the side effects of the primary active ingredient. This "buffer effect" is most clearly observed in the comparison between willow bark (*Salix* species) and isolated Aspirin (acetylsalicylic acid). Many believe willow is the natural source of aspirin, but willow bark contains only low quantities of the prodrug salicin, which is metabolized into various salicylate derivatives during absorption.[9]

Unlike synthetic Aspirin, willow bark does not damage the gastrointestinal mucosa.[15] The multi-component active principle of willow bark, including flavonoids and polyphenols, provides a broader mechanism of action than aspirin and is devoid of serious adverse events like GI bleeding.[9] In a comparative study using rats, an ethanol willow bark extract did not produce any mucosal lesions, whereas 9 of 10 rats treated with a corresponding dose of Aspirin showed significant stomach damage.[9] The omission of these natural buffers in the transition to synthetic Aspirin represents a fundamental betrayal of the "biological contract" to protect the organism while providing relief.[16]

Multi-Targeting: Orchestrated Signaling

While a synthetic drug like fluticasone targets a single inflammatory pathway, a plant "blend" can hit multiple receptors simultaneously. This pleiotropic effect allows for the stabilization of mast cells, the relaxation of smooth muscle, and the thinning of mucus through different but complementary chemical signals.[3] For instance, the phytochemicals in *Curcuma longa* (turmeric) act on $NF-\kappa B, STAT$, and $AP-1$ pathways, inhibiting lipoxygenases, cyclooxygenases, and phospholipases concurrently.[18] This "orchestrated signaling" respects the body's internal complexity, whereas the synthetic "isolated lever" approach treats the body as a broken machine that needs to be "fixed" with force.[19]

Bioavailability and the Whole Food Matrix

Natural complexes often include terpenes or lipids that help the body absorb medicine more efficiently. Bioactive compounds from plants are frequently accompanied by chemically related substances—"entourage compounds"—that appear inactive when administered individually but enhance the bioavailability of the primary active principle.[13] This is reflected in the whole food matrix, where nut fats stabilize antioxidants and polyphenols inhibit enzymes like α-amylase to slow carbohydrate digestion.[20]

The pharmaceutical attempt to replicate this natural efficiency leads to technologies like "ultramicronization." For instance, ultramicronized Palmitoylethanolamide (PEA-um) is designed to mimic the absorption rates that would occur naturally within a complex lipid matrix.[21]

Formulation Type	Bioavailability Profile	Mechanism
Naive PEA	Low (30-60% absorption)	Limited by large particle size and lipophilicity [21]
Ultramicronized PEA	High (82% absorption)	Increased surface area

		facilitates dissolution [21]
PEA with Natural Lipids	Very High (82% absorption)	Natural surfactants like saponins create emulsions [23]

The industry's push for ultramicronized isolates ignores the fact that nature already provides these delivery systems. Saponins from plants like *Quillaja saponaria* act as natural surfactants, producing water-soluble emulsions that increase the bioavailability of plant materials without the need for expensive industrial processing.[24]

The Audit of Omission: Respiratory Harmony and ASHMI

The most egregious example of "systemic violence" in respiratory medicine is the suppression of botanical synergies that match or exceed the power of synthetic steroids. The Anti-Asthma Herbal Medicine Intervention (ASHMI)—a formula containing *Ganoderma lucidum* (Ling-Zhi), *Sophora flavescens* (Ku Shen), and *Glycyrrhiza uralensis* (Gan Cao)—has been shown in human clinical trials to be a safe and effective alternative to prednisone for treating moderate-to-severe asthma.[3]

Clinical Reality vs. Standard of Care

In a double-blind trial involving 91 subjects, ASHMI produced improvements in lung function (FEV_1 and peak expiratory flow) that were comparable to prednisone.[3] More importantly, ASHMI had no adverse effects on adrenal function. While prednisone significantly decreased serum cortisol levels—an indicator of adrenal suppression—ASHMI significantly increased them.[3]

Metric	ASHMI (n=45)	Prednisone (n=46)	Result
FEV_1 Improvement	$P <$	$P <$	Comparable efficacy [3]
Symptom Score Reduction	Significant	Significant	No difference [3]
Adrenal	None	Significant (	ASHMI preserves homeostasis [3]

Suppression		$P <$)	
Cytokine Modulation	Balances Th1/Th2	General suppression	ASHMI is immunomodulatory [3]

Unlike corticosteroids that act as broad, non-specific immune suppressants, ASHMI acts on $Th2$ -driven inflammation to reduce airway hypersensitivity while increasing $IFN-\gamma$.[17] Animal models show that ASHMI treatment can completely reverse asthmatic characteristics for up to eight weeks post-therapy, suggesting a "life strategy" of re-education for the immune system rather than the "chemical hijack" of daily steroid use.[17] Despite these results being published in major journals like the *Journal of Allergy and Clinical Immunology* (JACI), ASHMI remains marginalized because its complex formula cannot be owned by a single entity.[3]

Endogenous Restoration: PEA and the PPAR-α Axis

The monetization of suffocation is further challenged by the existence of endogenous ligands like Palmitoylethanolamide (PEA). PEA is a signaling lipid produced "on demand" by the body in response to tissue damage and inflammation.[21] It manages the "suffocation" response by activating Peroxisome Proliferator-Activated Receptor Alpha ($PPAR-\alpha$), which downregulates $NF-\kappa B$ pathways and decreases the activation of the $NLRP3$ inflammasome.[12]

Bypassing the Synthetic Markup

While synthetic steroid inhalers carry a high markup and risk adrenal suppression, PEA works through the body's internal architecture to restore energy balance and WAT (White Adipose Tissue) homeostasis.[12] PEA has been shown to improve the incidence, duration, and severity of upper respiratory tract infections (URTIs) by modulating mast cells at sites of inflammation.[29] Clinical studies dating back to the 1970s demonstrated that daily supplementation with 1800 mg of PEA reduced fever and headaches, yet this endogenous "life strategy" is rarely discussed as a primary intervention.[29]

Mechanism	Synthetic Steroid Inhaler	Endogenous PEA Strategy
Target	Broad-spectrum immune suppression	$PPAR-\alpha$ and

		$TRPV$ signaling [28]
Adrenal Impact	Suppresses cortisol secretion [11]	Restores homeostasis "on demand" [21]
Side Effects	Adrenal burnout, dependency [3]	None reported; metabolic flexibility [12]
Delivery	Proprietary, patent-protected devices [4]	Oral lipids/whole food sources [20]

The system's intentional disregard for the $PPAR-\alpha$ receptor—a pathway the body already possesses to heal itself—illustrates the "betrayal of biological truth".[12] By focusing only on the synthetic "plug-in," the medical-industrial complex ensures a biological dependency that prevents the organism from utilizing its co-evolved signaling architecture.

Natural Synergies for Airway Harmony: Quercetin and Forskolin

Further evidence of a "truth perversion" in respiratory care is found in the comparison of natural mast-cell stabilizers and $cAMP$ boosters with their synthetic counterparts.

Quercetin: The Biological Original

Quercetin is a powerful flavonoid found in fruits and vegetables that acts as a natural mast-cell stabilizer.[31] In comparative studies, quercetin was found to be more effective than the synthetic drug cromolyn in blocking the release of pro-inflammatory cytokines ($IL-8$ and TNF) from human mast cells.[32] A crucial finding is that quercetin works prophylactically, while cromolyn rapidly loses its effect if not added exactly at the time of the trigger—a phenomenon known as tachyphylaxis.[32]

By inhibiting tryptase and histamine, quercetin protects connective tissue from degradation—a major concern for those with inflammatory or allergic conditions.[31] When paired with bromelain, an enzyme from pineapple, the bioavailability of quercetin is enhanced, and mucus is broken down safely without the chemical side effects of synthetic mucolytics.[31]

Coleus Forskohlii and cAMP

Synthetic rescue inhalers function by increasing $cAMP$ to open airways. *Coleus forskohlii*, a root used in Ayurvedic medicine, contains forskolin, which directly activates the adenylate

cyclase enzyme to elevate $cAMP$ levels.[34] This mechanism is identical to the end-result of synthetic inhalers but avoids the β-receptor desensitization that occurs with chronic use of synthetic agonists.[35] Despite its clinical value for asthma, glaucoma, and heart failure, the pharmaceutical industry focuses primarily on the extraction and purification of forskolin for high-cost products rather than encouraging the use of the whole-plant material.[36]

The Monetization of Inflammation: IBD and the Biologics Trap

The "systemic violence" of withholding non-patentable medicine is perhaps most visible in the treatment of Inflammatory Bowel Disease (IBD). Biologics like Humira (adalimumab) and Remicade (infliximab) are the "gold standard" for inducing remission in ulcerative colitis and Crohn's disease, yet they are among the most expensive drugs in existence, costing between $\$15,000$ and $\$30,000$ per year per person.[38]

The Cost of Omission

Both turmeric (*Curcuma longa*) and frankincense (*Boswellia serrata*) have demonstrated clinical efficacy in IBD that matches standard-of-care drugs like mesalazine.[18] A non-inferiority trial comparing *Boswellia serrata* extract H15 to mesalazine in active Crohn's disease patients showed that the herbal extract was not only equally effective but also superior in terms of a benefit-risk evaluation.[40]

Treatment	Annual Drug Cost	Adverse Event Profile
Humira (Adalimumab)	$$19,249 - $\$22,21$	Serious infections, black label warning [38]
Entyvio (Vedolizumab)	$$21,385 - $\$26,32$	Infusion reactions, immunosuppression [38]
Curcumin/Boswellia	$<$ (OTC)	Safe, no significant side effects [18]

Curcumin blocks $TNF-\alpha$ production and action in both animal models and humans, yet it is rarely offered as a first-line therapy because it is "very inexpensive" and cannot be protected by the same patent layers as monoclonal antibodies.[39] Adalimumab-treated patients have a significantly higher risk of serious infection requiring hospitalization (HR 5.11) compared to those on older biologics, yet the system continues to push these high-markup, high-risk

"chemical hijacks" over natural alternatives.[43]

Enforced Fragility and the De-Evolution of the Organism

The "death machine" doesn't just damage the body; it creates a biological "chemical tether." By suppressing natural immune communication (like the mast-cell or $Th2$ response) instead of modulating it, the system makes the organism unable to survive without the synthetic plug-in.[3] This de-evolution is evident in the transition from synergistic plant medicine to isolated molecules:

1. **Severing Symbiosis**: Replacing co-evolved "life strategies" with isolated molecules that the body does not recognize as medicine, but as a hijack.
2. **Spiritual and Evolutionary Erasure**: Treating the body as a broken machine to be "fixed" with a single patented lever, rather than a self-correcting system designed for homeostasis.
3. **Biological Debt**: The resulting side effects are not just "unfortunate accidents"; they are the interest paid on the biological debt incurred by overriding natural signaling.

The persistence of the synthetic model, despite 98% of these interventions being replaceable by natural alternatives that respect the biological contract, is a form of "monetized suffocation".[7] By gatekeeping the "natural originals"—which are safer and often more effective—the FDA and the pharmaceutical industry essentially hold the breath and health of the population hostage for profit.[4]

Conclusions: Dismantling the Industrial Default

To dismantle a system that prioritizes patents over pathogens, a new "Biology First" framework is required. This would involve identifying conditions where a safe, natural original (like ASHMI or PEA) matches the synthetic plug-in and documenting how the industrial model suppresses that data through gatekeeping.

Restructuring the dialogue around health requires:

- A "Patent vs. Potency" audit to expose where highly effective natural strategies have been ignored because they could not be captured via IP.[1]
- A "Synergy-Only" lens that focuses exclusively on the entourage effect—how multiple compounds work in concert to achieve homeostasis without adrenal burnout or chemical dependency.[13]
- An "Audit of Omission" that identifies the structural betrayal of providing a "gold standard" that prioritizes corporate earnings over the fundamental right to breathe and heal using the natural originals co-evolved with our species.

The "systemic violence" of the current medical model is not merely a failure of science but a crime of information. By stripping away the industrial "safety" narrative and exposing the lethality of the default, the organism can begin to reclaim its power and return to a state of self-regulated homeostasis.[10]

Works cited

1. Botanical Drugs - Emery Pharma, accessed March 5, 2026, https://emerypharma.com/solutions/botanical-drug-development/
2. Botanical Drug Development Guidance | Dalton, accessed March 5, 2026, https://www.dalton.com/Content/files/botanical-drug-development-guidance.pdf
3. Efficacy and tolerability of anti-asthma herbal medicine intervention ..., accessed March 5, 2026, https://pubmed.ncbi.nlm.nih.gov/16159618/
4. Tertiary Patents on Drugs Approved by the FDA - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12761334/
5. What is a Botanical Drug? | FDA, accessed March 5, 2026, https://www.fda.gov/about-fda/cder-offices-and-divisions/what-botanical-drug
6. Botanical Drug Review - FDA, accessed March 5, 2026, https://www.fda.gov/media/94221/download
7. The Double-Edged Sword of Pharmaceutical Patents: A Roadmap for Policy Reform, accessed March 5, 2026, https://www.drugpatentwatch.com/blog/the-double-edged-sword-of-pharmaceutical-patents-a-roadmap-for-policy-reform/
8. From Patent to Patient: FDA Policy Ideas to Protect and Speed Medical Innovation, accessed March 5, 2026, https://www.jdsupra.com/legalnews/from-patent-to-patient-fda-policy-ideas-5391413/
9. Willow Species and Aspirin: Different Mechanism of Actions - ResearchGate, accessed March 5, 2026, https://www.researchgate.net/publication/49746890_Willow_Species_and_Aspirin_Different_Mechanism_of_Actions
10. Xiu-Min Li - Icahn School of Medicine at Mount Sinai, accessed March 5, 2026, https://scholars.mssm.edu/en/persons/xiu-min-li
11. Adrenal suppression by inhaled corticosteroids in patients with asthma: A systematic review and quantitative analysis - PubMed, accessed March 5, 2026, https://pubmed.ncbi.nlm.nih.gov/26831841/
12. Palmitoylethanolamide Promotes White-to-Beige Conversion and Metabolic Reprogramming of Adipocytes: Contribution of PPAR-α - MDPI, accessed March 5, 2026, https://www.mdpi.com/1999-4923/14/2/338
13. The Entourage Effect in Cannabis Medicinal Products: A Comprehensive Review - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11870048/
14. Understanding the Entourage Effect: Why Whole-Plant Medicine Matters | MARY & MAIN, accessed March 5, 2026,

https://maryandmain.com/what-is-the-entourage-effect/
15. Willow species and aspirin: different mechanism of actions - PubMed, accessed March 5, 2026, https://pubmed.ncbi.nlm.nih.gov/21226125/
16. Willow bark | Uses, side-effects - Arthritis UK, accessed March 5, 2026, https://www.arthritis-uk.org/information-and-support/understanding-arthritis/arthritis-treatments/complementary-and-alternative-treatments/types-of-complementary-treatments/willow-bark/
17. ASHMI™ induced long-lasting tolerance to allergen exposure in an asthma model is interferon - γ, but not TGF-β dependent - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2946973/
18. Effects of Boswellia Serrata Roxb. and Curcuma longa L. in an In ..., accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6316569/
19. The effectiveness and safety of curcumin as a complementary therapy in inflammatory bowel disease: A protocol of systematic review and meta-analysis - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7581072/
20. The Power of Synergy and Bioavailability in the Whole Food Matrix - WholisticMatters, accessed March 5, 2026, https://wholisticmatters.com/power-of-synergy-bioavailability-whole-food-matrix/
21. Oral Ultramicronized Palmitoylethanolamide: Plasma and Tissue Levels and Spinal Anti-hyperalgesic Effect - Frontiers, accessed March 5, 2026, https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2018.00249/full
22. Micronized/ultramicronized palmitoylethanolamide displays superior oral efficacy compared to nonmicronized palmitoylethanolamide in a rat model of inflammatory pain - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4171547/
23. Palmitoylethanolamide as a Supplement: The Importance of Dose-Dependent Effects for Improving Nervous Tissue Health in an In Vitro Model - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11354262/
24. CA3128653A1 - Nanoemulsion compositions comprising saponins for increasing bioavailability - Google Patents, accessed March 5, 2026, https://patents.google.com/patent/CA3128653A1/en
25. (PDF) Perspectives on Saponins: Food Functionality and Applications - ResearchGate, accessed March 5, 2026, https://www.researchgate.net/publication/373570419_Perspectives_on_Saponins_Food_Functionality_and_Applications
26. Safety and Tolerability of an Antiasthma Herbal Formula (ASHMI™) in Adult Subjects with Asthma: A Randomized, Double-Blinded, Placebo-Controlled, Dose-Escalation Phase I Study - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2830609/
27. The anti-asthma herbal medicine ASHMI acutely inhibits airway smooth muscle contraction via prostaglandin E2 activation of EP2/EP4 receptors - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3882537/
28. Role of PPAR-α, PPAR-γ, and PPAR-δ in palmitoylethanolamide-induced... -

ResearchGate, accessed March 5, 2026, https://www.researchgate.net/figure/Role-of-PPAR-a-PPAR-g-and-PPAR-d-in-palmitoylethanolamide-induced-inhibition-of_fig4_235395130
29. The Efficacy of Palmitoylethanolamide (Levagen+) on the Incidence and Symptoms of Upper Respiratory Tract Infection—A Double Blind, Randomised, Placebo-Controlled Trial - MDPI, accessed March 5, 2026, https://www.mdpi.com/2072-6643/15/20/4453
30. What is PEA? - Zero Pain Philosophy, accessed March 5, 2026, https://www.zeropainphilosophy.com/post/what-is-pea
31. Quercetin, Mast Cell Activation Syndrome, and Histamine Intolerance | The EDS Clinic, accessed March 5, 2026, https://www.eds.clinic/articles/quercetin-mast-cell-activation-syndrome-and-histamine-intolerance
32. Quercetin Is More Effective than Cromolyn in Blocking Human Mast ..., accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3314669/
33. Quercetin and Mast Cell Stabilization - The Fascia Institute, accessed March 5, 2026, https://fasciainstitute.org/quercetin-mast-cell-stabilization/
34. Total biosynthesis of the cyclic AMP booster forskolin from Coleus forskohlii - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5388535/
35. applied sciences - OPUS at UTS, accessed March 5, 2026, https://opus.lib.uts.edu.au/bitstream/10453/137301/1/applsci-09-04089-v3.pdf
36. Total biosynthesis of the cyclic AMP booster forskolin from Coleus forskohlii | eLife, accessed March 5, 2026, https://elifesciences.org/articles/23001
37. Biotechnological interventions for the production of forskolin, an active compound from the medicinal plant, Coleus forskohlii - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11016037/
38. Cost Comparison - Pharmacoeconomic Review Report: adalimumab (Humira) - NCBI - NIH, accessed March 5, 2026, https://www.ncbi.nlm.nih.gov/books/NBK538746/
39. Curcumin: an orally bioavailable blocker of TNF and other pro-inflammatory biomarkers, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3753829/
40. [Therapy of active Crohn disease with Boswellia serrata extract H 15] - PubMed, accessed March 5, 2026, https://pubmed.ncbi.nlm.nih.gov/11215357/
41. Therapy of active Crohn disease with Boswellia serrata extract H 15]. - ResearchGate, accessed March 5, 2026, https://www.researchgate.net/publication/12116722_Therapy_of_active_Crohn_disease_with_Boswellia_serrata_extract_H_15
42. Curcumin for the clinical treatment of inflammatory bowel diseases: a systematic review and meta-analysis of placebo-controlled randomized clinical trials - PMC, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11973083/
43. Comparison of Infliximab and Adalimumab in Biologic-Naive Patients With Ulcerative Colitis: A Nationwide Danish Cohort Study - PubMed, accessed March 5, 2026, https://pubmed.ncbi.nlm.nih.gov/27913244/

44. Infliximab versus Adalimumab, Which One Is Better for Ulcerative Colitis? - PMC - NIH, accessed March 5, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7960970/
45. A long-overlooked class of medicines—botanical drugs—may finally gain traction in the U.S., as scientists and the FDA forge new pathways to bring plant-based therapies to mainstream medicine. | Journal of Integrative Dermatology, accessed March 5, 2026, https://jintegrativederm.org/doi/10.64550/joid.ysj2js31
46. FDA Advances a "Plausible Mechanism" Framework for Rare Disease Drug Development and Signals a Shift to a Single Trial with Confirmatory Evidence Default for All Drugs | Advisories | Arnold & Porter, accessed March 5, 2026, https://www.arnoldporter.com/en/perspectives/advisories/2026/02/fda-advances-a-plausible-mechanism-framework-for-rare-disease-drug-development-and-shifts-to